A Case Study on Physics-Informed Neural Networks for Option Pricing

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Abstract

Physics-Informed Neural Networks (PINNs) combine neural networks and partial differential equations (PDEs) by enforcing that the neural approximation satisfies the underlying PDE. In quantitative finance, this promises a unified differentiable framework for forward pricing. This paper presents a case study comparing PINNs against classical finite-difference methods for the 1D Black-Scholes equation. We show that a PINN trained with SiLU activations and adaptive sampling mitigates the spectral bias of standard architectures, reducing the RMSE compared to a Tanh baseline. We also explore Sobolev distillation as a surrogate for Greek calculation. Finally, we deploy the trained model to C++ via ONNX Runtime, achieving a 121x speedup in inference latency compared to a Crank-Nicolson solver that solves the PDE from scratch. We provide an honest discussion of the limitations, including the lack of a multi-parameter benchmark and the trade-offs between accuracy and speed.

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1 Mathematical Setting and The Black-Scholes PDE

1.1 The Probabilistic Framework

We consider a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions, where $\mathbb{P}$ is the physical (historical) measure. The market consists of a risk-free money market account B with constant interest rate r, and a risky asset S following a Geometric Brownian Motion (GBM). Under $\mathbb{P}$, the dynamics are:

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{P}}, \tag{1}$$

where $\mu \in \mathbb{R}$ is the drift, $\sigma > 0$ is the volatility, and $W^{\mathbb{P}}$ is a standard $\mathbb{P}$-Wiener process.

Theorem 1.1 (Itô's Lemma)**.** *Let $f(t, S)$ be a twice continuously differentiable function in S and once in t. Then:*

$$\mathrm{d}f(t, S_t) = \left(\frac{\partial f}{\partial t} + \mu S_t \frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 f}{\partial S^2} \right) \mathrm{d}t + \sigma S_t \frac{\partial f}{\partial S} \,\mathrm{d}W_t^{\mathbb{P}}. \tag{2}$$

1.2 Risk-Neutral Measure and Feynman-Kac

By the Fundamental Theorem of Asset Pricing, the absence of arbitrage is equivalent to the existence of an equivalent martingale measure $\mathbb{Q} \sim \mathbb{P}$. By Girsanov's Theorem, the Radon-Nikodym derivative $\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}$ defines a new Wiener process $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \frac{\mu - r}{\sigma} t$. Under $\mathbb{Q}$, the discounted asset price is a martingale, and the dynamics become:

$$\mathrm{d}S_t = r S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t^{\mathbb{Q}}. \tag{3}$$

Theorem 1.2 (Feynman-Kac)**.** *Let $V(t, S)$ be the no-arbitrage price of a European contingent claim with payoff $\Phi(S_T)$. Then, under the risk-neutral measure $\mathbb{Q}$:*

$$V(t, S) = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t)} \Phi(S_T) \mid S_t = S \right]. \tag{4}$$

Moreover, $V(t, S)$ is the unique solution to the Black-Scholes PDE.

1.3 Time-to-Maturity Formulation

For numerical tractability, we define the time-to-maturity $\tau = T - t$. Let $\tilde{V}(\tau, S) = V(T - \tau, S)$. The Black-Scholes PDE becomes a forward-in-time parabolic equation:

$$\frac{\partial \tilde{V}}{\partial \tau} = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \tilde{V}}{\partial S^2} + rS \frac{\partial \tilde{V}}{\partial S} - r\tilde{V}, \tag{5}$$

subject to the initial condition $\tilde{V}(0, S) = \Phi(S)$. For a European call, the terminal payoff is $\Phi(S) = \max(S - K, 0)$.

Definition 1.3 (PDE Residual)**.** We define the nonlinear differential operator $\mathcal{R}$ applied to a function $V(\tau, S)$ as:

$$\mathcal{R}[V](\tau, S) := \frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS \frac{\partial V}{\partial S} + rV. \tag{6}$$

The exact solution satisfies $\mathcal{R}[V] \equiv 0$ on the entire domain.

2 Numerical Baseline: Crank-Nicolson Scheme

To establish a rigorous baseline, we discretize eq. (5) using the Crank-Nicolson scheme. We define a uniform spatial grid $S_i = i\Delta S$ for $i = 0, \ldots, M$ and a temporal grid $\tau_n = n\Delta\tau$ for $n = 0, \ldots, N_t$.

2.1 Discretization and Matrix Formulation

Applying central differences for the spatial derivatives and averaging the implicit and explicit Euler schemes, we obtain:

$$\frac{V_i^{n+1} - V_i^n}{\Delta\tau} = \frac{1}{2}\left(\mathcal{L}_h V_i^{n+1} + \mathcal{L}_h V_i^n\right), \tag{7}$$

where $\mathcal{L}_h$ is the discretized differential operator:

$$\mathcal{L}_h V_i = a_i V_{i-1} + b_i V_i + c_i V_{i+1}, \tag{8}$$

with coefficients:

$$a_i = \frac{\sigma^2 S_i^2}{2\Delta S^2} - \frac{r S_i}{2\Delta S}, \tag{9}$$

$$b_i = -\frac{\sigma^2 S_i^2}{\Delta S^2} - r, \tag{10}$$

$$c_i = \frac{\sigma^2 S_i^2}{2\Delta S^2} + \frac{r S_i}{2\Delta S}. \tag{11}$$

This yields a sequence of tridiagonal linear systems:

$$\underbrace{\left(I - \frac{\Delta\tau}{2} L_h\right)}_{\mathcal{A}} \mathbf{V}^{n+1} = \underbrace{\left(I + \frac{\Delta\tau}{2} L_h\right)}_{\mathcal{B}} \mathbf{V}^n + \mathbf{b}, \tag{12}$$

where $\mathbf{b}$ accounts for boundary conditions. The matrix $\mathcal{A}$ is tridiagonal, time-independent, and strictly diagonally dominant. Thus, it is invertible and stable for any choice of $\Delta\tau$.

2.2 The Thomas Algorithm

Because $\mathcal{A}$ is tridiagonal, we solve eq. (12) using the Thomas algorithm (a simplified Gaussian elimination), which reduces the computational complexity from $\mathcal{O}(M^3)$ to $\mathcal{O}(M)$.

Proposition 2.1 (Complexity)**.** *The Thomas algorithm solves eq. (12) in $\mathcal{O}(M)$ operations per time step. The total complexity of the Crank-Nicolson solver is $\mathcal{O}(MN_t)$.*

2.3 Convergence Analysis

The truncation error of the Crank-Nicolson scheme is $\mathcal{O}(\Delta\tau^2 + \Delta S^2)$. However, the payoff $(S-K)^+$ has a discontinuous first derivative (a kink) at $S = K$, which theoretically degrades the spatial convergence to $\mathcal{O}(\Delta S)$ near the strike. As shown in table 1, the observed convergence ratio is ≈ 4 (i.e., $\mathcal{O}(N^{-2})$),

Table 1: Convergence of the C++ Crank-Nicolson solver against the analytical Black-Scholes price ($S_0 = K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.2$).

$M = N_t$	C_{CN}	$\lvert C_{\text{CN}} - C_{\text{BS}}\rvert$	Ratio
50	10.18992825	2.607×10^{-1}	–
100	10.38812450	6.246×10^{-2}	4.17
200	10.43510760	1.548×10^{-2}	4.03
400	10.44672294	3.861×10^{-3}	4.01
800	10.44961893	9.646×10^{-4}	4.00

meaning the kink does not contaminate the global error order. The solver also strictly preserves the Put-Call parity (error $< 10^{-8}$) and arbitrage bounds.

3 Physics-Informed Neural Networks: Architecture and Training

We approximate the solution map $\tilde{V} : [0, T] \times (0, S_{\max}) \to \mathbb{R}$ using a feedforward neural network with 4 hidden layers and 64 neurons per layer (approx. 12,000 parameters). The network is trained by minimizing the composite loss:

$$\mathcal{L}(\theta) = \lambda_{\mathcal{R}}\mathcal{L}_{\mathcal{R}}(\theta) + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}}(\theta) + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}}(\theta), \tag{13}$$

where $\mathcal{L}_{\mathcal{R}}$ penalizes the PDE residual. The boundary conditions are enforced via a soft penalty in the loss ($\lambda_{\text{BC}} = 10$), not as hard constraints.

3.1 Spectral Bias and Architectural Mitigations

A well-documented pathology of standard neural networks is *spectral bias*: they learn low-frequency components much faster than high-frequency ones. The option payoff has a kink at $S = K$, making the PDE residual singular at $\tau = 0$. To mitigate this, we employ:

1. **SiLU Activation:** $f(x) = x \cdot \sigma(x)$, which has non-vanishing gradients for large positive inputs.
2. **Adaptive Sampling:** 50% of points are uniform, 50% follow $\mathcal{N}(K, 5)$ near the strike.
3. **Gradient Clipping:** Max norm of 1.0 to prevent instability.

3.2 Sobolev Distillation for Greeks

We train a variant using *Sobolev training* [2], penalizing the L^2 distance between the network's spatial derivative and the analytical Black-Scholes Delta. **We explicitly acknowledge that this is supervised learning (distillation).** A pure PINN learns solely from the PDE. Here, we demonstrate that the network can act as a fast surrogate for both prices and Greeks. This is an engineering choice, not a pure PDE solution.

3.3 Experimental Results

Table 2 reports the results for the different architectures. The RMSE and Max Error are computed on a test grid $S \in [0.5K, 1.5K]$, $\tau \in (0, T)$. The **Pure Tanh** baseline is included to demonstrate the mitigation of spectral bias. The **Pure SiLU** model achieves the best pricing RMSE and is deployed in C++.

Table 2: Performance of PINN architectures against the analytical Black-Scholes solution.

Model	Epochs	Price RMSE	Max Error	PDE Res.	$V(T, K)$
Black-Scholes (ref.)	–	0	0	0	10.45058
PINN Pure (tanh)	20,000	1.82×10^{-1}	7.81×10^{-1}	0.65	10.533
PINN Pure (SiLU)	20,000	$\mathbf{4.22 \times 10^{-2}}$	1.99×10^{-1}	0.13	10.367
PINN Sobolev (SiLU)	10,000	6.70×10^{-2}	2.68×10^{-1}	0.15	10.546

Remark 3.1 (Trade-offs and Limitations)**.** The SiLU activation clearly mitigates the spectral bias compared to the Tanh baseline, reducing the Price RMSE by a factor of 4. The Sobolev-distilled model achieves a competitive price but was trained for fewer epochs. However, we note that **we do not report a dedicated Greeks error table in this draft**. The claim that Sobolev improves Greek accuracy is based on the loss convergence, not on a formal error metric. Furthermore, this is a single-parameter case study; a multi-strike/multi-maturity benchmark is left for future work.

4 C++ Deployment and Latency Benchmark

We exported the **Pure (SiLU)** model to ONNX (opset 11) and executed it in C++ using ONNX Runtime (v1.17). We benchmarked the inference latency on a single CPU core (Intel i7), averaging over 1,000 forward passes.

Table 3: Inference latency. The Crank-Nicolson solver computes a 400x400 grid from scratch.

Method	Latency (μs)	Price ($S = 100, \tau = T$)	Abs. Error
Black-Scholes (analytic)	< 1	10.45058	0 (ref.)
Crank-Nicolson (C++)	655	10.44672	3.86×10^{-3}
PINN Pure (C++ / ONNX)	**5.40**	10.37	8.29×10^{-2}

Remark 4.1 (Contextualizing the Speedup and Training Cost)**.** The PINN achieves a 121x speedup over a naive Crank-Nicolson solver. However, for the standard 1D Black-Scholes equation, the analytical formula is the fastest option. The PINN's advantage emerges as a surrogate model for high-dimensional or local volatility PDEs where analytical formulas do not exist. We also note that the PINN requires an offline training phase (approx. 20 minutes on a GPU for 20,000 epochs). This upfront cost must be amortized over millions of real-time inferences, which is typical in high-frequency trading or real-time risk management.

5 Conclusion

In this case study, we evaluated Physics-Informed Neural Networks for option pricing. We demonstrated that SiLU activations mitigate the spectral bias of standard neural networks, reducing the pricing RMSE. We also explored Sobolev distillation for Greek calculation and deployed the model to C++ via ONNX Runtime, achieving a 121x speedup compared to a numerical PDE solver.

This proves that PINNs are viable surrogate models for numerical PDE solvers. While the analytical formula remains superior for 1D Black-Scholes, PINNs show great promise for high-dimensional and inverse problems. Future work will focus on a multi-parameter benchmark (varying strikes, maturities, and seeds), formal Greek error metrics, and the inverse calibration of local volatility surfaces.

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